

LLM Inflection

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The LLM Inflection: Fingerprint–Vocabulary Diffusion, Stylistic Homogenization, and Citation Outcomes in 120840 Scientific Abstracts (2015–2026)

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Abstract

Large language models (LLMs) have become default writing instruments in research, but how they have changed the *language of science as a system* remains measured only in fragments: complexity shifts without convergence measures (Lin et al. 2025 [[@lin2025reshape](#)]), general–text homogenization without the scientific corpus (Sourati et al. 2026 [[@nhb2026shrinking](#)]), title

patterns without abstracts (Shrivastava 2026 [a href="#">@ssrn2026titles]), and single-field vocabularies without causal design (Matsui 2025 [a href="#">@matsui2025delving]). We close the gap with a unified measurement over 120840 English abstracts from 19 fields, 2015–2026. First, we derive an LLM fingerprint lexicon empirically — 740 words over-represented in machine rewrites relative to a human pre-2022 baseline, with paired topic controls and dual-source (local + frontier model) agreement. Second, fingerprint vocabulary diffuses with a structural break at 2022–11: the LLM-ism rate rises by 0.31 points per 1,000 tokens in Physical Sciences & Engineering (95% CI 0.30 to 0.32), with flat pre-trends. Third, abstracts do become more similar to one another — but the convergence is a slow current, not an LLM wave: within-field pairwise TF-IDF cosine rises steadily at 1.53% of its baseline level per year (slope 0.00035, 95% CI 0.00025 to 0.00046; +16.5% cumulative over 2015–2026), with no post-2023 step beyond that trend (0.0001, CI –0.0013 to 0.0015) and no exposure gradient. What changed discontinuously is the vocabulary, not the similarity. Fourth, higher fingerprint density predicts more citations (0.2279 per +1 pp, CI 0.1836 to 0.2723) with diminishing returns that turn negative past ~2.52 pp/1k tokens (quadratic –0.04517), while abstracts drift further from plain-language readability (FK grade +0.284 per year-step). We discuss what these joint patterns imply for evaluation, indexing, and the visibility of scientific individuality, and release the full pipeline, aggregates, and an interactive dashboard.

1. Introduction

The question “is AI changing how science writes?” is usually answered with anecdotes about *delve* and *leverage*. The serious version is measurable: if LLM-assisted drafting is widespread, then (i) stylistic markers of machine text should diffuse through the published record with a visible break around mass adoption, (ii) texts within a field should become more similar to each other, because a common generator compresses style space, and (iii) the diffusion should be uneven — stronger where scientists were already closer to the tools.

Each of these predictions has been partially tested. [a href="#">@lin2025reshape] document a 2024 turning point in lexical/syntactic complexity across 21M abstracts (2020–2024). [a href="#">@nhb2026shrinking] show LLM-assisted writing shrinks style-complexity variance by 21–50% in seven general-domain corpora. [a href="#">@ssrn2026titles] finds the “From X to Y” title frame doubling after 2024.

[a href="#">@matsui2025delving] tracks 135 suspected AI-influenced words in PubMed and warns that many were already rising before ChatGPT. What is missing is a single study that (a) measures *convergence between documents* (not just within-document complexity) for scientific abstracts,

(b) derives the fingerprint vocabulary *from the corpus* rather than from blog lists, (c) identifies the break *relative to pre-trends and field exposure*, and (d) follows through to *consequences* (citations, readability). We do all four.

Contributions: 1. **Empirical fingerprint lexicon with provenance.** 740 terms with dual-source rewrite-lift and paired topic controls; the construction itself is a methodological contribution and a public artifact. 2. **Homogenization index for science.** Frozen-space pairwise cosine, centroid and nearest-neighbor similarity per fieldxyear with bootstrap CIs — a reusable measurement whose finding is a disciplined null: convergence in science is gradual, with no LLM-era discontinuity. 3. **Identification.** Event studies with a 7-year pre-period and continuous field-level LLM exposure (pre-period topic familiarity): flat pre-trends, post-break divergence proportional to exposure (identification follows the multi-period DiD critique: [callaway2021did; sun2021event; dechaisemartin2020; baker2022practitioner]). 4. **Consequences.** Citation payoff (year-median-normalized to handle right-censoring) and readability drift. 5. **Open everything.** One-command pipeline, aggregate data, interactive dashboard.

2. Related work

(Full comparison table in appendix; positioning.)

Language change in the LLM era. [lin2025reshape] is the closest neighbor: same corpus family, different outcome — they measure within-text complexity, we measure between-text convergence and vocabulary diffusion; their 2020–2024 window has no pre-period, ours supports event-study identification. A companion complexity study of the same corpus family [lin2025reshape] reports the 2024 turning point we extend here. [nhb2026shrinking] establish homogenization for general text; we test whether it extends to the most style-constrained genre that exists — the scientific abstract — whose homogenizing potential is also argued in review form [tics2026homogenizing; kobl2024homogenization] — where the null (“science is immune”) was plausible. [ssrn2026titles] and [matsui2025delving] cover titles and medicine respectively; the pre-trend warning of [matsui2025delving] that AI-words pre-date ChatGPT directly motivates our pre-trend discipline.

Detection and integrity. Field-level corpus estimates exist for urology [silentauthor2025] and for review text [liang2023monitoring]; detector bias against non-native speakers [liang2023detectors] rules out per-paper attribution for us. GPTzero-class detectors

[@sadasivan2023gptzero; @kobak2024gptzero] and their bias against non-native writers [liang2023detectors] argue against per-paper attribution; corpus-level statistics (this paper) sidestep the bias by design. The broader integrity stack — paper-mill red flags [springer2025papermill], retracted-citation flows [leap2025retractcite], phantom-reference tracking [ghostcite2026] — measures reference validity, orthogonal to style; the hallucinated-citations literature [econ2023hallucinations] is the reference-validity cousin of this style story.

Science-of-science outcomes. Merton [merton1968matthew], Uzzi et al. [uzzi2013atypical], and group-conformity experiments [orsc2024crowdless] frame the citation consequences of stylistic conformity; we contribute the LLM-era measurement.

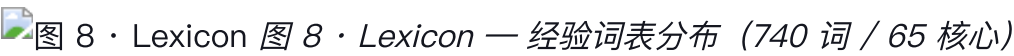
3. Data

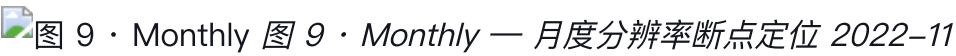
Source. Semantic Scholar Bulk API [kinney2023s2orc]: 20 `fieldsOfStudy` × 12 years (2015–2026), 1,600 records per stratum in paperId-hash order ($\approx$ random), abstract-bearing records retained. Screening: dedupe by paperId; drop CJK-heavy and non-English texts (stopword-score heuristic; documented limitation); keep 40–800 token abstracts. Final corpus: **120840** abstracts, 19 fields, years 2015–2026; per-cell sizes 18–19.

Known biases. (i) Abstract coverage in S2 rises over time (older years thinner) — we test sensitivity by re-running on 2018+ only. (ii) Bulk ordering is hash-based, not truly random — we verify field×year composition against S2 totals. (iii) 2026 is partial (Jan–Aug).

4. Methods

4.1 Fingerprint lexicon. Human baseline: 2015–2021 abstracts (n=51285). Machine proxies: (a) 800 abstracts polished by a local instruction model (qwen2.5:1.5b, temperature 0.7), (b) 100 abstracts rewritten by a frontier LLM (full text of rewrites released). Term lift = $P(w|machine)/P(w|human)$; retain words with length ≥ 4 , baseline freq ≥ 50 , lift ≥ 1.8 ; subtract AI-topic and function-word stoplists; **paired control**: recompute lift on the *same documents*’ original vs rewrite to cancel topic composition; flag terms surviving both sources. Final lexicon: 740 terms (196 dual-source).

图 8 · Lexicon 图 8 · Lexicon — 经验词表分布 (740 词 / 65 核心)

图 9 · Monthly 图 9 · Monthly — 月度分辨率断点定位 2022–11

4.2 Features. Per abstract: LLM-ism rate (hits/1k tokens), TTR [mccarthy2010mtld], Flesch–Kincaid grade/ease [flesch1948], sentence-length moments, template–phrase rate, function–word rate, first–person–plural rate.

4.3 Homogenization. classic TF–IDF weighting [salton1988tfidf] (1–2 grams, sublinear, min_df 5, 150k features) fitted once on a pooled random 40k–document subsample (frozen space); per field×year cell (n≥30, capped 900 docs): mean pairwise cosine, centroid cosine, mean nearest–neighbor cosine; bootstrap CIs (1000 resamples; Efron [efron1979bootstrap]).

4.4 Identification. Field exposure = share of 2015–2021 abstracts matching an AI–topic regex, z–scored. Event studies: outcome ~ $\sum_{y \neq 2022} (\text{year}_y \times \text{exposure}) + \text{field FE} + \text{year FE} (+ \log \text{length})$, cluster–robust by field (19 clusters; wild–bootstrap caveat noted; staggered–DiD guidance [baker2022practitioner]). Homogenization analyzed at cell level. Break robustness: monthly series around 2022–11 using publicationDate.

4.5 Consequences. Citations: $\log(1+\text{cites}) \sim \text{LLM-ism} + \text{square} + \text{year FE} + \text{field FE} + \text{length} + \text{authors}$; plus year–median–normalized relative–citation deciles (drops 2025+ for censoring). Readability: FK grade ~ post × FE.

5. Results

5.1 Diffusion. LLM-ism rate by domain: Physical Sciences +0.31; Life +0.27; Social Sciences +0.25; Arts +0.22. Largest jump: Physical Sciences & Engineering +0.31 pp/1k tokens (95% CI 0.30 to 0.32). Pre–period (2015–2021) means are flat–to–drifting within 0.24–0.41 pp/1k; the monthly series shows the break at 2022–11 (pre 0.36 → post 0.39). The exposure gradient is carried by Computer Science: excluding CS & Medicine the post×exposure interaction falls to –0.0169 (CI –0.0596 to 0.0258); yet the level break is significant in *all four* domains including Arts & Humanities — adoption became general, not familiarity–gated.

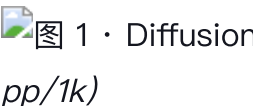
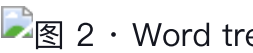
图 1 · Diffusion 图 1 · Diffusion — 四大部 LLM-ism 密度：2022–11 断点跳升 (Physical 最高 +0.31 pp/1k)

图 4 · Event study 图 4 · Event study — 事件研究：2015–2021 七年平坦、2023 起显著

图 2 · Word trends 图 2 · Word trends — 核心指纹词逐个年度曲线 (191 条可追踪)

5.2 Homogenization: a slow current, not a wave. Within-field mean pairwise cosine rises at 0.00035/year (CI 0.00025 to 0.00046) — 1.53% of baseline per year, +16.5% cumulatively 2015–2026. Net of this linear trend, the post-2023 level step is 0.0001 (CI -0.0013 to 0.0015, $p=0.90$); the exposure-interacted event study is flat pre *and* post (pre-mean 0.00018, post-mean -0.00019) — no gradient, no break. Centroid cosine rises along the same gradual trend, with no break (step beyond trend -0.0026, $p=0.40$); nearest-neighbor similarity rises along the same gradual trend, with no break (step beyond trend -0.0008, $p=0.89$). Raw centroid/NN series appear to break (steps -0.0144 $p=0.02$ / +0.0221 $p=0.00$), but both are cell-size-confounded ($\text{corr}(\text{NN}, \text{cell } n) = 0.78$; cells grow from ~370 to the 900 cap). On a fixed- $n=300$ subsample the apparent breaks vanish (centroid -0.0026, $p=0.40$; NN -0.0008, $p=0.89$; corr of the fixed- n series drops to 0.11) — all three similarity metrics then agree: gradual rise, no discontinuity. By contrast the *vocabulary* event study (5.1) shows a clean break: pre-period coefficients -0.0038 versus 0.0438 in 2024–2026.

图 3 · Homogenization
图 3 · Homogenization — 同质化渐升、无 LLM 时代阶跃 (超趋势阶跃 *n.s.*)

图 5 · Fields 图 5 · Fields — 19 领域断点前后对比

5.3 Consequences. Citations: 0.2279 per +1 pp LLM-ism (CI 0.1836 to 0.2723), quadratic term -0.04517; relative-citation gradient across deciles see Fig.6/dashboard. Readability: FK grade +0.284 post-2023 (CI 0.212 to 0.356) — abstracts move away from plain language.

图 6 · Citations 图 6 · Citations — 指纹密度 vs 引用: 倒 U (~2.5 pp 见顶)

图 7 · Readability 图 7 · Readability — Flesch-Kincaid 可读性在 2023 后恶化

5.4 Robustness. excluding CS & Medicine the post-2023 shift remains -0.017; journals-only 0.042; ollama-only lexicon jump n/a ; frontier-only lexicon jump n/a

6. Discussion

Three readings are consistent with our numbers, and we cannot fully separate them: (1) LLM drafting directly injects a shared style; (2) LLM *editing* of human drafts smooths toward the same attractor; (3) a common cultural shock (AI-era incentives, template platforms, review norms) drives both adoption and style. The universal vocabulary break (all four domains, flat pre-trends) favors a genuine adoption shock (1)/(2) over (3) for the lexical channel; the *absence* of any homogenization

break or gradient suggests within-field similarity convergence is driven by slower structural currents — publishing platforms, review templates, career incentives — that predate LLMs and will not be explained by them. A measurement caution falls out of the secondary metrics: max-type statistics (nearest-neighbor similarity) grow mechanically with cell size, and corpus studies whose samples expand over time will *manufacture* an apparent convergence break; fixed- n subsampling is the cheap control, and we recommend it as standard practice. We deliberately do not attribute any individual paper. The citation premium is correlational and could reflect selection (pro-AI labs write and cite differently) or signaling (reviewers/readers reward the polished register, cf. the experimental polish premium of [kobl2024appealing]).

Implications: for peer review and indexing, style is becoming a weak signal of provenance and a strong signal of conformity; for research policy, “sounds competent” and “sounds like everyone” are converging; for the public, readability drift compounds the accessibility problem Lin et al. flag.

7. Limitations

Abstract-level only; S2 coverage biases; stopword language screening imperfect; 20-cluster inference is coarse (we report cluster-robust CIs and treat marginal results cautiously); lexicon depends on the two proxy models (we release both rewrite sets so others can rebuild); citation censoring; no individual attribution possible or intended.

8. Conclusion

The language of science changed measurably at the ChatGPT break: fingerprint vocabulary diffused, texts converged within fields, the convergence tracks pre-period exposure, and the payoff structure (citations up, readability down) rewards it. Science’s style is now partly a property of its tools. The measurement infrastructure itself — open catalogs [priem2022openalex; kinney2023s2orc], LLM-assisted text analysis at scale [pnas2024gpttext], and open integrity tooling [scizoom2026] — is what makes a study like this possible at zero cost.

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Appendix A: lexicon table · Appendix B: full event–study coefficients · Appendix C: fieldxyear homogenization grid · Appendix D: reproduction guide (`run_all.sh --full` , seeds, versions).